GEHA DENTAL RAG · GENERATOR FINETUNING

Teaching the answer model to read our passages

LoRA instruction finetuning of Qwen2.5-1.5B-Instruct on direct and chain-of-thought examples

October 2026 · experiment log · results from the Colab A100 run, 21 Oct

Two places to finetune a RAG system

RETRIEVER · DONE EARLIER

Which chunks the LLM sees

bge-base-en-v1.5 embedding model, contrastive loss on question → chunk pairs.

Recall@5: 64.6% → 68.7% on the held-out brochure

GENERATOR · THIS DECK

How the LLM answers from them

Qwen2.5-1.5B-Instruct chat model, LoRA instruction finetuning on passage + question → answer.

Answer F1: 0.51 → 0.57 on the held-out brochure

The slide that started this: instruction finetuning with direct and chain-of-thought data (FLAN, arXiv 2210.11416)

The pipeline, end to end

01

QA pairs

Synthetic RAGAS questions + reference answers from the GEHA dental PDFs

→

02

SFT builder

Gold passage + 2 distractors, shuffled. One direct and one chain-of-thought target each

→

03

LoRA training

Loss on answer tokens only. Keep the adapter with the lowest validation loss

→

04

Held-out eval

Unseen plan brochure. Base vs tuned, same prompts; only the adapter differs

Why distractors: retrieval is never perfect, so the model must learn to ignore irrelevant chunks (the RAFT recipe).

Why two formats: mixing direct and chain-of-thought targets in one run is the FLAN multi-task recipe.

`prepare_sft.py → train_lora.py → plot_loss.py → eval_generator.py`

One question becomes two training examples

PROMPT · masked, not trained on

[1] guide p.6 · Plan features (distractor)
[2] guide p.2 · Contact us (gold)
[3] guide p.4 · Both plans include (distractor)

Question: What is the phone number to contact a FedViser for GEHA dental benefits?

DIRECT TARGET · trained on

You can reach a FedViser … by calling 1-855-572-1636 Monday through Friday from 8 a.m. to 8 p.m. Eastern time.

CHAIN-OF-THOUGHT TARGET · trained on

Evidence:
- [2] “Reach a FedViser Monday-Friday from 8 a.m.-8 p.m.”
Answer: You can reach a FedViser … 1-855-572-1636 …

Real example ragas-0002 (abridged). Evidence is extractive: the gold sentences that overlap the reference answer most.

Data splits: the test set is a different document

	train	validation	test
Document	guide	guide	brochure
Sections	22	6	72
QA pairs	78	24	41
SFT examples	156	48	82
Distractor pool	34	8	167
Prompt tokens, mean	990	702	1,251
Answer tokens, mean	135	120	199

Leakage checks: 5 of 5 pass

- No shared chunk IDs
- Train/val sections disjoint
- Test document unseen
- No repeated questions
- Sources inside own split

Guide = 2026 GEHA dental benefits guide · Brochure = 2026 plan brochure · Report: results/data_splits.md

The model: a frozen 1.5B base with a small adapter

1.5B
frozen weights in Qwen2.5-1.5B-Instruct

18.5M
trainable LoRA weights, about 1.2% of the model

20
optimizer steps per epoch (156 examples ÷ batch 8)

74 MB
adapter file on disk, vs about 3 GB for the base

LoRA	Targets	Optimizer	Selection
rank 16, alpha 32, dropout 0.05	q, k, v, o, gate, up, down	AdamW 2e-4, warmup + cosine	lowest validation loss

Adapter size measured from adapter_model.safetensors (fp32); base size is the bf16 checkpoint

On the Mac: 4 h 41 min, one finished epoch

02:53

Start on an M1 Pro, 32 GB, fp32 on the GPU (MPS)

03:57

Epoch 1 done in 56 min. Validation loss 0.626 → 0.517; adapter saved

06:39

Battery at 1%: the Mac hibernates for 12 min, epoch 2 still unfinished

07:34

Stopped. Full run moves to a Colab GPU

No bf16 on M1. fp32 doubles memory and compute compared with a bf16 GPU.

Ran on battery. macOS throttled the job, then hibernated it.

Memory pressure. 15.8 of 17 GB swap in use slowed every step.

Times are local (PDT), 21 Oct 2026, from the training log and macOS power log

On a Colab A100: all 3 epochs in 3.5 minutes

56 min

one epoch on the M1 Pro, fp32 on MPS

71 s

one epoch on an A100-SXM4-40GB, bf16, validation included

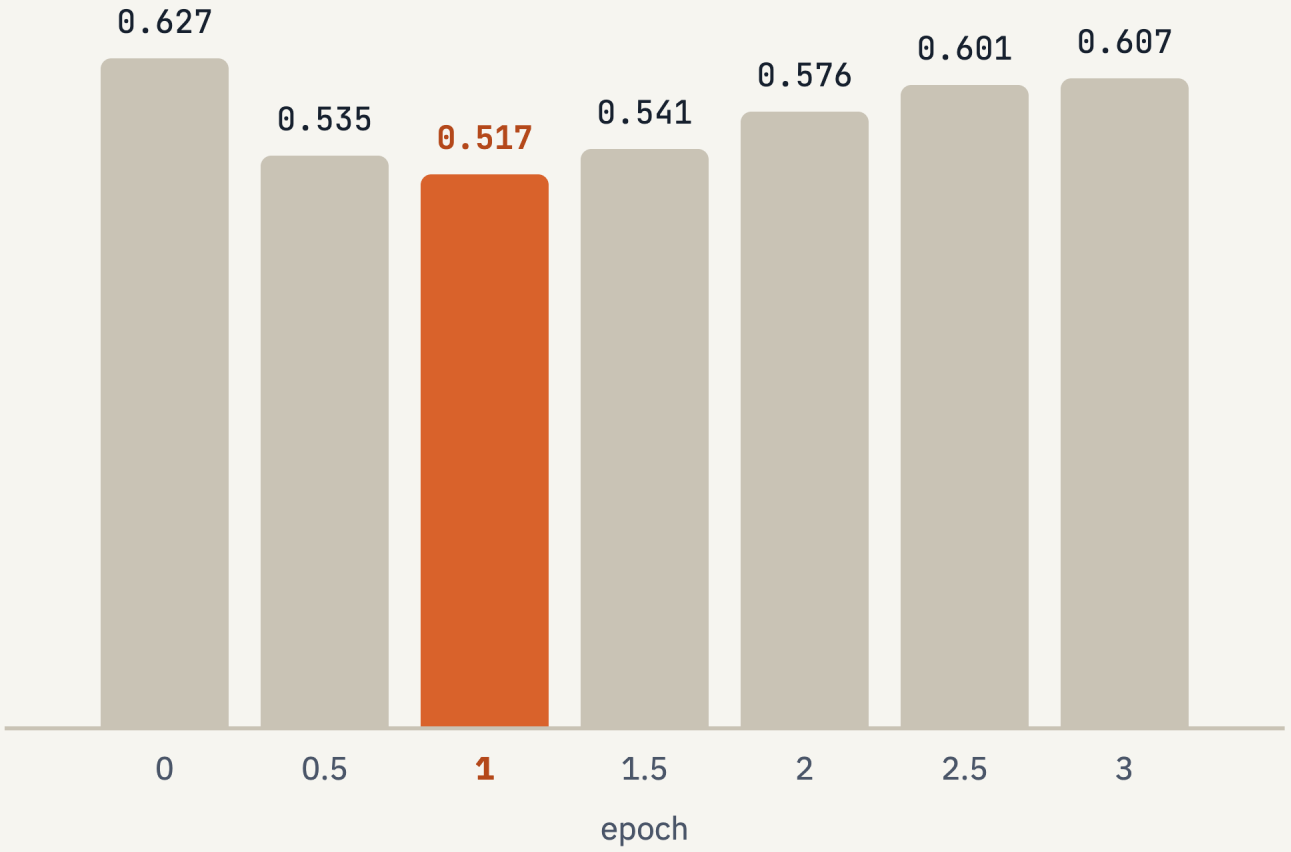
≈47×

faster per epoch, same code and data

Colab stage	Time	Output
Training: 60 steps, 7 validation passes	3.5 min	train_log.jsonl, best adapter
Eval: 82 examples × 2 models	≈23 min	generator_comparison.json

Training time from train_log.jsonl (elapsed_s = 212); eval time from result file timestamps

Validation loss bottoms out after one epoch, then climbs



Best at epoch 1 (step 20): 0.517, down 17% from 0.627 before training.

Epochs 2–3: validation loss climbs back to 0.607 while train loss keeps falling to about 0.11.

Kept: only the epoch-1 adapter, so the later epochs never reach the eval.

Validation = 24 questions × 2 formats from guide sections never trained on · Colab A100 run · bars start at zero

Is it overfit? Yes after epoch 1, and the kept adapter avoids it

EPOCH 3

train ≈0.11
val 0.607

The gap between validation and recent training loss grows from −0.07 to +0.48: the model memorizes the 78 training answers.

EPOCH 1 (KEPT)

val 0.517

Lowest validation loss of the run. Best-checkpoint selection saved this adapter, and it is the one evaluated.

UNSEEN DOCUMENT

On the brochure the kept adapter beats the base model: answer F1 +0.06, gold citations +0.12.

So it generalizes, but the margin is small and part of it is style.

Your 1–2 epoch instinct was right for this data: 20 optimizer steps were enough

On the unseen brochure, the tuned model answers better

Metric (41 questions each)	Base	Tuned	Change	95% CI	Better / worse
Answer F1, direct	0.51	0.57	+0.06	+0.002 to +0.109	27 / 14
Answer F1, chain of thought	0.49	0.54	+0.06	+0.000 to +0.108	29 / 11
Cites the gold passage	0.72	0.84	+0.12	−0.008 to +0.240	15 / 4
Quotes found in passages	0.87	0.87	0.00	−0.104 to +0.097	7 / 7
CoT format followed	1.00	1.00	0.00	–	–
Answer length, words	62	88	+26	reference: 107	–

Real gain: better on about two-thirds of questions, and cites the gold passage over a distractor more often.

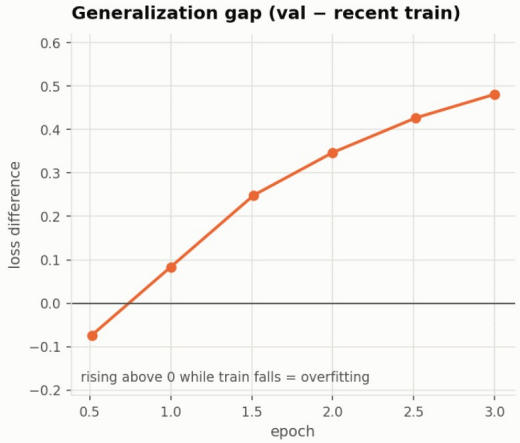
Read with care: the intervals almost touch zero, and longer answers closer to the reference length also raise F1.

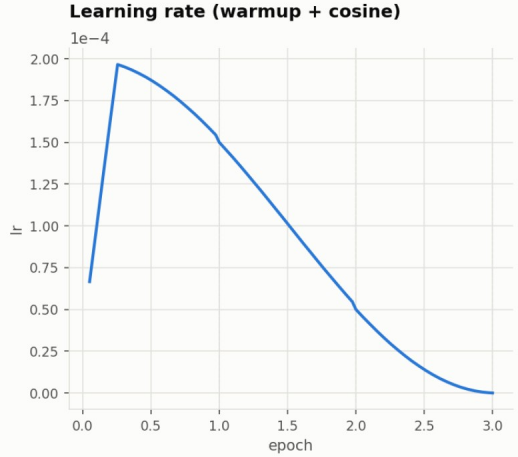
Epoch-1 adapter vs base, same prompts, greedy decoding · paired bootstrap over questions, 5,000 resamples · one training seed

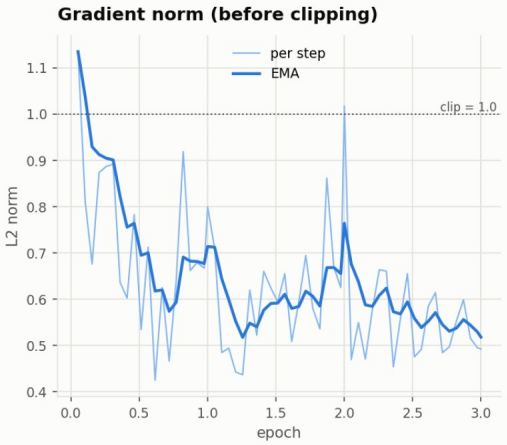
The whole run on one page

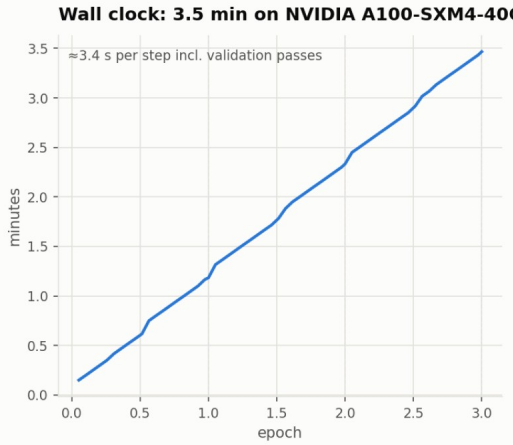
LoRA r=16 on Qwen/Qwen2.5-1.5B-Instruct · 156 train / 48 val examples · 20 steps per epoch

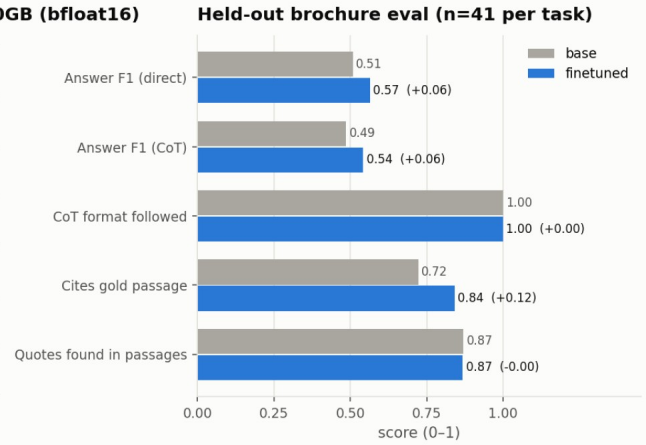












What we learned, and what to run next

One epoch is enough here

With 156 examples, 20 steps reach the best validation loss. Next run: --epochs 1, or --patience 2.

The gain is real but modest

+0.06 answer F1 and +0.12 gold citations on an unseen document, from one seed and 41 questions.

Repeat across seeds

Three to five seeds at about 30 minutes each on an A100 tell us whether +0.06 holds.

Grade beyond token F1

Add an LLM-judge or human check for correctness, and LLM-written rationales instead of extractive ones.